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Palaeomagnetism of the Permo-Triassic Araguainha impact structure (Central Brazil) and implications for Pangean reconstructions

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SUMMARY

The configuration of the Pangea supercontinent has been a topic of intense debate for almost half a century, a controversy that stems from discrepancies between the geology-based Pangea-A and the palaeomagnetically based Pangea-B. Recent palaeomagnetic compilations aimed at resolving this controversy have identified the poor quality of palaeomagnetic data from Gondwana for Permian times as a major obstacle. Specifically, the vast majority of Gondwanan poles come from sedimentary rocks that are prone to biases from compaction or are poorly dated. Here, we present a new palaeomagnetic pole for cratonic South America based on impact melts from the 254.7 ± 2.5 Ma Araguainha impact structure (AIS). The granite basement, the impact-generated melt sheet and veins were sampled at 28 sites (169 specimens) and provided a reliable palaeomagnetic record similar to that of volcanic rocks. Alternating field and thermal demagnetization indicate a stable characteristic remanent magnetization carried by both magnetite and haematite. All sites but one show a single palaeomagnetic direction of normal polarity with a mean direction of Dec = 357.4°; Inc = −38.9°; $N = 28$; $k = 62.35$; $\alpha_{95} = 3.5°$, yielding a palaeomagnetic pole (AIS) at Lat = −84.2; Lon = 326.6; $K = 83.5$; $A_{95} = 3.6°$; $S_B = 9.6°$. The new pole provides a firm constraint on the position of Gondwana which is consistent with the Pangea A configuration.

Key words: Palaeomagnetic secular variation; Palaeomagnetism applied to tectonics; Impact phenomena.

INTRODUCTION

The relative position of southern and northern parts of Pangea (Fig. 1) in the late Palaeozoic has been a topic of intense debate for almost half a century, ever since Ted Irving showed a dramatic inconsistency between Wegener's original palaeogeography (Pangea-A) and the then-available pre-Jurassic palaeomagnetic data (Irving 1977). Strict interpretation of the palaeomagnetic data for the late Palaeozoic requires a ∼1500 km overlap between Gondwana and Laurussia; Gondwana is too far North and/or Laurussia is too far South. To resolve this overlap, Irving proposed a new palaeogeography (Pangea-B) where Gondwana is longitudinally displaced relative to Laurussia, placing, for example, NW South America next to SE North America in the late Palaeozoic. This configuration would require a huge dextral displacement across the Laurussia–Gondwana boundary during the Triassic to accommodate the transition from the Pangea-B to the classical Pangea-A (e.g. Irving 1977; Hallam 1983), which is well-constrained from Jurassic palaeomagnetic data as well as from marine magnetic anomalies from the central Atlantic Ocean. More recently, palaeomagnetic studies by Muttoni *et al.* (1996) suggested that the transition from Pangea-B to Pangea-A was completed much earlier, in the Middle Permian, coeval with the opening of the Neo-Tethys Ocean. The authors based their interpretation on a set of poles from sedimentary and volcanic units from Southern Alps, which were taken as a proxy for West Gondwana because of its very poor palaeomagnetic database. Angular differences between this West Gondwana apparent polar wander (APW) path and that of Laurussia would reach minimum values for a Pangea-B in the early Permian, and for a Pangea-A after the middle Triassic. In the late Permian to middle Triassic, the angular distance between APW paths for all Pangea reconstructions (Pangea-A and Pangea-B) is similar. After that, Muttoni *et al.* (2003, 2009) have confirmed their interpretation based on volcanic rocks from the Southern Alps and from lateritic weathering profiles of Permian age from northern Iran and western Karakoram, Pakistan.

Palaeomagnetic data for the heart of Western Gondwana—Africa and South America—are still scant. Recent compilations highlight the poor quality of Carboniferous to Triassic palaeomagnetic record for Gondwana; this deficiency being particularly acute for latest Permian to early Triassic times (e.g. Brandt *et al.* 2009; Domeier *et al.* 2012; Torsvik *et al.* 2012). Most of the Permo-Triassic palaeomagnetic data are from South America sedimentary rocks, which are susceptible to palaeomagnetic biases caused by sediment compaction, remagnetization, as well as age uncertainty (e.g. Bilardello

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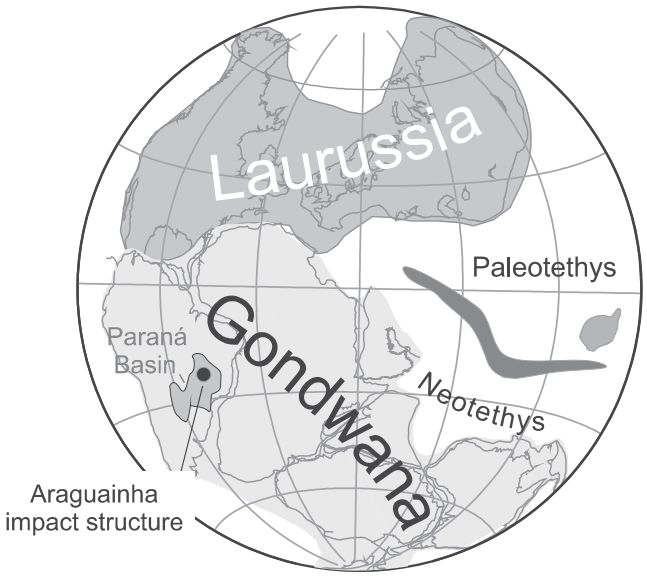


Figure 1. Location of the Araguainha crater (circle) and outline of the Paraná basin into the Pangea A palaeogeography. Pangea-A configuration rotated using the palaeomagnetic pole of this study, euler angles from Bullard *et al.* (1965).

& Kodama 2010; Font *et al.* 2012). Igneous rocks are the best palaeomagnetic targets for establishing reference poles, given their high-fidelity record of the geomagnetic field and amenability to geochronological techniques. However, volcanic rocks from the stable interior of Gondwana are scarce for this time period.

In order to provide better palaeomagnetic constraints for the West Gondwana in the Permo-Triassic, we studied the impact-related materials of the Araguainha impact structure (AIS), which was recently dated by Tohver *et al.* (2012). These impact-related rocks are composed of sedimentary rocks, impact melts and granitic rocks (e.g. Engelhardt *et al.* 1992; Machado *et al.* 2009). Our study focused on the palaeomagnetic characteristics of the impact melts and the granitic target rocks.

GEOLOGICAL SETTING AND SAMPLING RATIONALE

The Araguainha impact event affected the sedimentary rocks of the northern Paraná Basin and its basement in central Brazil (16°47′S, 52°59′W). The 40-km-diameter structure (Fig. 1) is delimited by concentric faults, annular rings and a 10-km-wide central uplift of exhumed ca. 512 Ma granite core surrounded by a collar of sedimentary rocks of the Paraná Group (Lana *et al.* 2007, 2008; Tohver *et al.* 2012; Tohver *et al.* 2013; Fig. 2a). Impact-related melts are observed in contact with the granite (Engelhardt *et al.* 1992; Machado *et al.* 2009; Yokoyama *et al.* 2012), both in the form of veins that crosscut the porphyritic granite, and small bodies of melt sheet with polymict breccia deposits on top of the melt sheet (Fig. 2b). A recent geochronological investigation using U-Pb SHRIMP and $^{40}Ar/^{39}Ar$ dating of neocrystallized phases in the melt rock provides a precise age for the impact at 254.7 ± 2.5 Ma, coinciding with the Permo-Triassic limit (Tohver *et al.* 2012).

Impact melts are generated after the passage of the initial compressional shock wave, with temperatures in the target rocks increasing above the dry melting point of most rock-forming minerals, producing a large volume of melt (e.g. Grieve & Cintala 1992; Dressler & Reimold 2001). A portion of melt is ejected from the transient cavity, and the remaining melts are redistributed along the modified impact crater floor in the form of a melt sheet and dykes

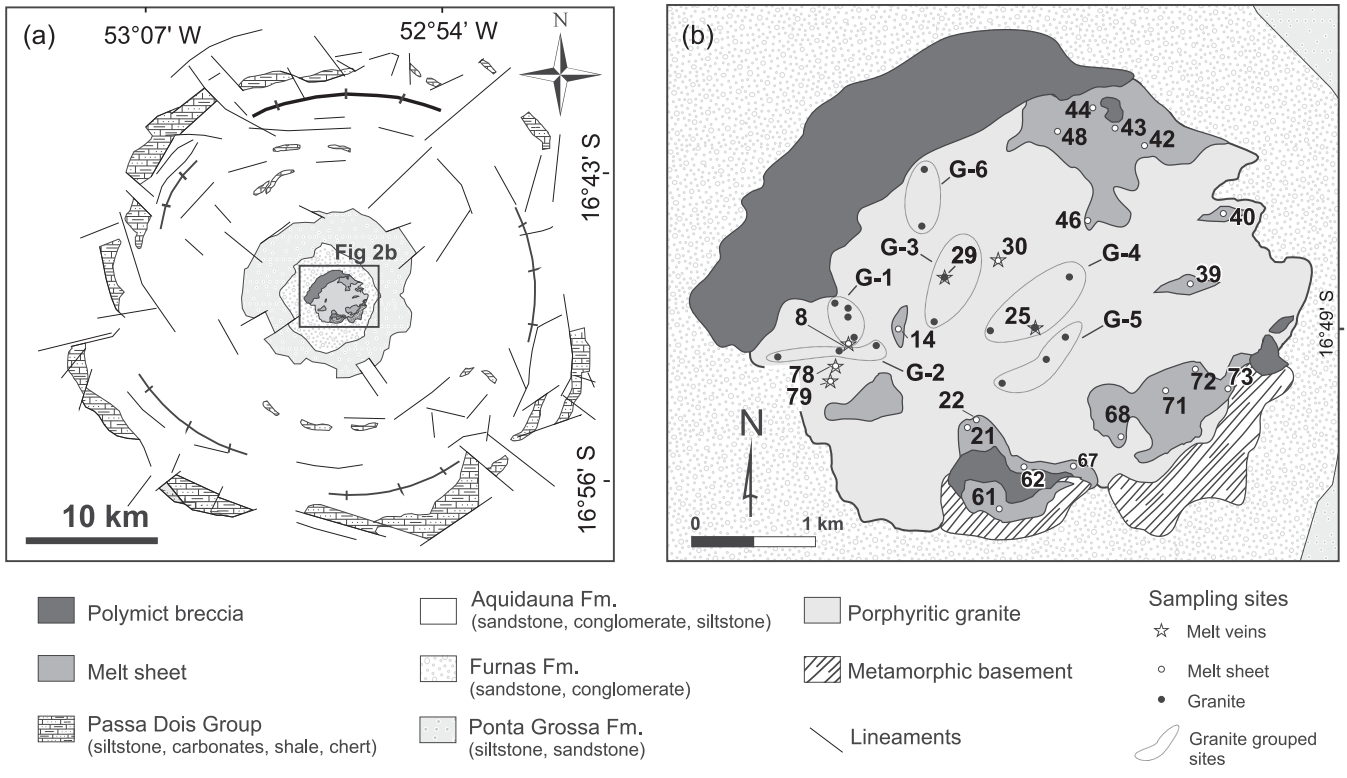


Figure 2. Geological maps of Araguainha: (a) simplified geological map of the impact structure and (b) detail of the central uplift area and location of sampling sites.

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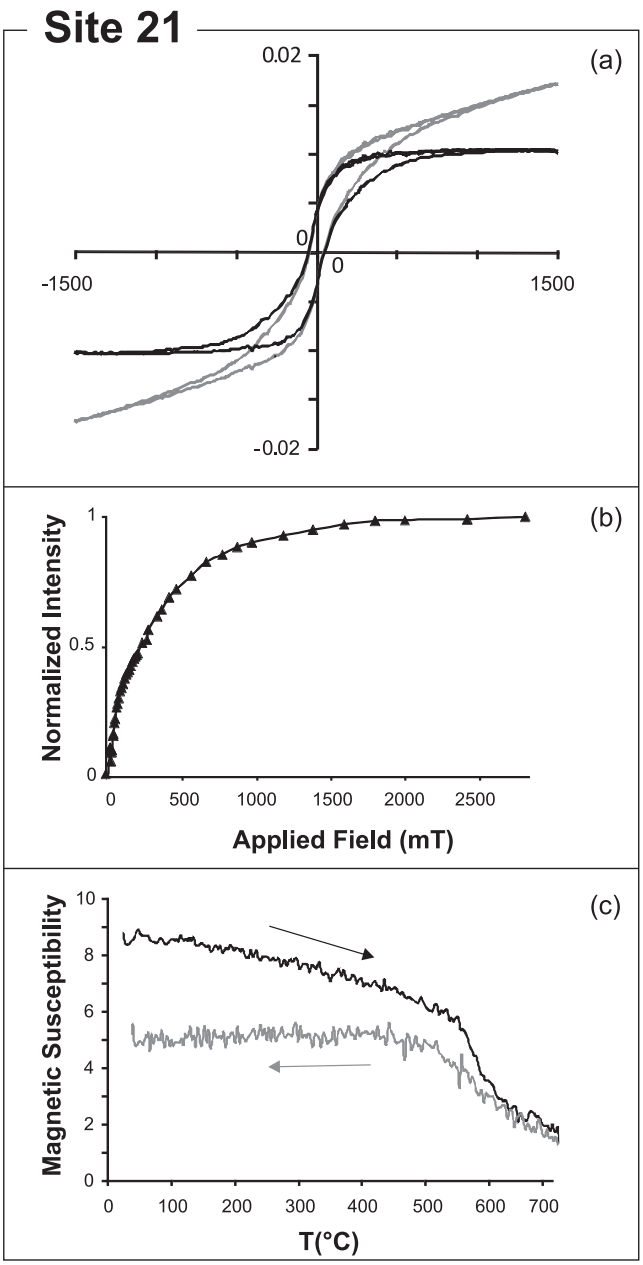


Figure 3. Magnetic mineralogy of Araguainha impact melts (site 21), showing the co-occurrence of low- and high-coercivity magnetic phases: (a) hysteresis loops; (b) thermomagnetic curves and (c) isothermal magnetization acquisitions.

(e.g. Dressler & Reimold 2001; Spray 2010). The magnetic remanence in autochthonous impact melts is of thermal origin, similar to magmatic rocks (e.g. Pesonen *et al.* 1992; Carporzen *et al.* 2005; Salminen *et al.* 2009; Koch *et al.* 2012; Eitel *et al.* 2014), and is acquired during cooling over a time frame that is proportional to the volume of the melt body.

One of the basic requirements in palaeomagnetic studies is that site-based directions average out the secular variation (SV) of the geomagnetic field. For this reason, a comprehensive sampling through a long period of time (10^4–10^5 yr) is necessary. In impact melt rocks, as with volcanic rocks, the remanent magnetization is acquired during cooling through the unblocking temperature of magnetic minerals. An analogue is provided by the 200-m-thick melt sheet at the 65 km Manicouagan impact structure of Quebec, Canada. A thermal model by Onorato *et al.* (1978) considered the latent heat of crystallization as well as conductive heat loss. At high temperatures (ca. 1000 °C), cooling rates vary as the inverse square of the distance from the melt sheet boundary, but cooling rates vary more linearly as cooling proceeds to the Curie temperature range of haematite and magnetite, 680 and 578 °C, respectively. The calculations by Onorato *et al.* (1978) show that the centre of the Manicouagan melt sheet would have reached 630 °C between 3300 and 5300 yr after the impact. However, these analytical calculations of conductive cooling ignore the heightened permeability of the fractured basement, which will serve as a locus for enhanced, long-lived hydrothermal circulation. For example, a recent thermochronological study of the smaller, 23 km Lappajärvi crater suggests elevated temperatures (230–410 °C) in the central uplift over 0.6–1.6 Ma (Schmieder & Jourdan 2013).

The thickness of the impact melt sheet can be directly estimated from the size of the impact structure (e.g. Daubar & Kring 2001). Based on these scalar relations, we estimated a thickness of 400 m for the melt sheet of the 40-km-wide AIS. The present-day erosional level left only the basal 50 m of this melt sheet (e.g. Engelhardt *et al.* 1992; Machado *et al.* 2009), which is consistent with a 250–350 m depth of erosion estimated by Lana *et al.* (2007). Taking into account the original melt sheet thickness, the cooling time from an initial temperature of 1450 °C (Machado *et al.* 2009) down to the Curie temperature of magnetite 580 °C is ∼3200 yr. The incipient hydrothermal system formed in the Araguainha (e.g. Engelhardt *et al.* 1992; Jovane *et al.* 2011) likely contributed to increase the timing of remanence acquisition by increasing cooling time, but also by inducing chemical and thermochemical magnetization in the melt and the host granite.

Given the analytical and empirical constraints that indicate cooling over a possible interval of 10^3–10^5 yr, we propose that magnetization acquisition in widely distributed volumes of Araguainha melt rock and its granitic basement is sufficiently long-lived to average out SV. Our detailed sampling covers different sectors and all stratigraphic units of the impact melt sheet, the melt veins and the granitic host rocks, in order to capture as much of the cooling history of the impact structure as possible.

METHODS

A total of 28 sampling sites composed of 17 sites in the melt sheet, five sites in melt veins and six sites in the granite were studied (Fig. 2b). We sampled three to nine oriented samples from each sampling site. For the granite, the sites enumerated from G1 to G6 consist of a clustering of three to six neighbouring sampling sites of Yokoyama *et al.* (2012). Standard 2.2 × 2.5 cm cylindrical specimens cut from individual samples were subjected to AF and/or thermal demagnetization. Demagnetization was carried out using 15–20 progressive steps, up to 160 mT and 700 °C, respectively. AF demagnetization was performed in either a three-axis AF-demagnetizer coupled with a 2G Enterprises SQUID magnetometer or a LDA AGICO tumbler demagnetizer (AGICO, Czech Republic). Thermal demagnetization was carried out with an ASC oven (peak temperatures within ±2 °C, the samples remain at peak temperatures for 20 min). Remanent magnetizations were measured with a SQUID magnetometer (model 755UC, 2G) or a spinner magnetometer (model JR6A, Agico). These instruments are housed in a magnetically shielded room (ambient field <1000 nT) at the Paleomagnetic Laboratory of the University of São Paulo (USP). Characteristic remanent magnetizations (ChRMs) were calculated

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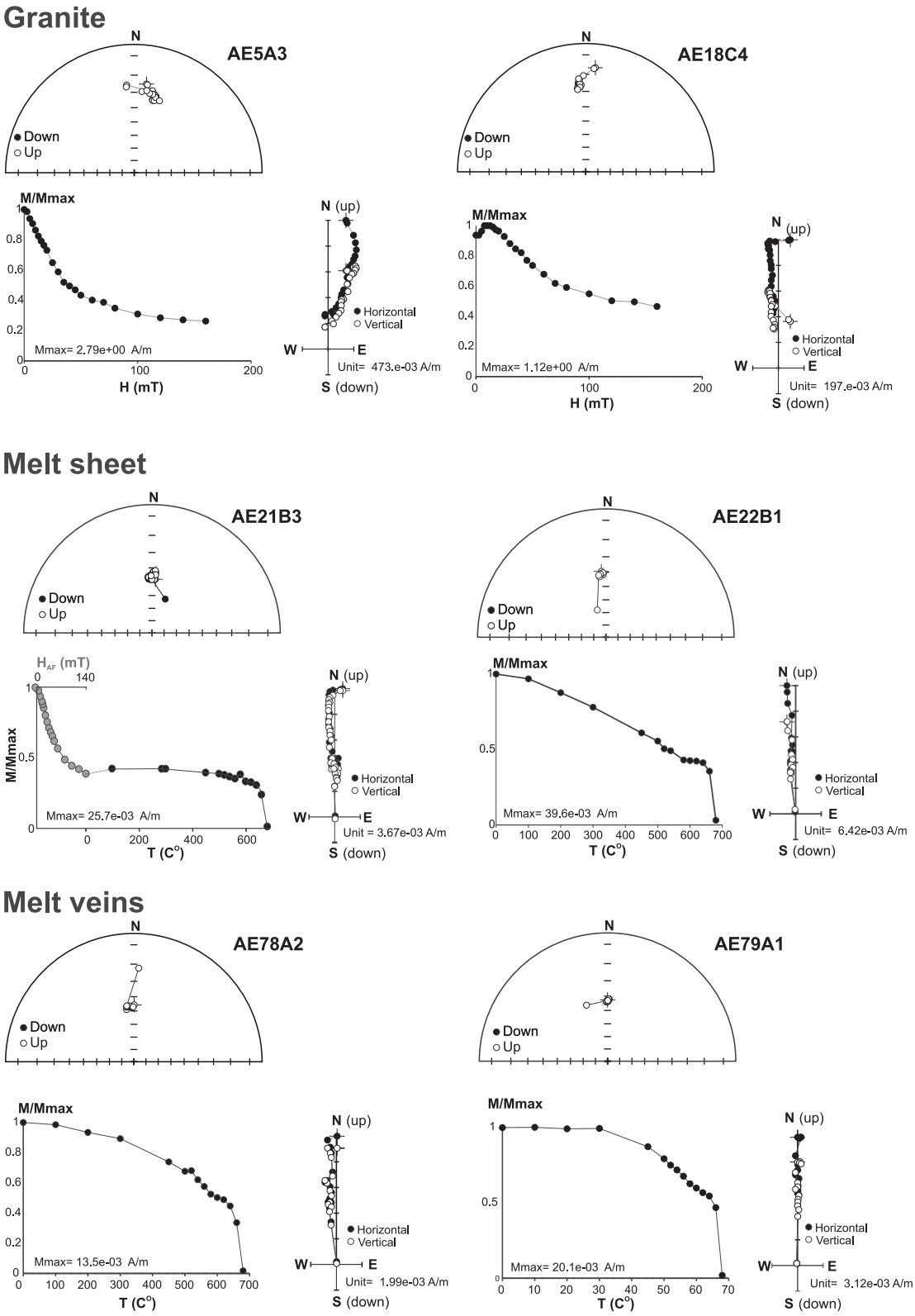


Figure 4. Examples of AF and thermal demagnetizations (stereographic projections, orthogonal projections and magnetization intensity decay curves) for melt sheets and melt veins samples.

by principal component analysis (PCA; Kirschvink 1980) or great circles analyses (Halls 1976). Mean directions were obtained by Fisherian statistics or by the routine of McFadden & McElhinny (1988) to combine planes and directions. The method of variable cut-off of Vandamme (1994) was used to select the VGPs for the final mean.

Magnetic mineralogy was characterized in representative samples using hysteresis cycles, isothermal remanent magnetization

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Table 1. Palaeomagnetic results for the Araguainha impact structure.

		Mean characteristic directions				VGP		
Site	n N^{-1}	Declination (deg)	Inclination (deg)	(deg)	k	Plat (deg S)	Plon (deg W)	Swi (deg)
*8	12/13	355.7	−40.5	3.9	126.1	−82.5	338.8	8.3
14	3/6	5.3	−27	15.4	65.1	−84.2	193.1	8.9
21	4/4	358.1	−41.8	3.1	900	−82.5	320.6	2.7
22	6/6	1.9	−44.4	6.2	133.5	−80.4	296.7	5.7
*29	7/8	9	−38.3	2.3	667.2	−80.3	247.5	3.3
*30	6/9	1.3	−41.1	4	285.4	−83.1	297.6	5.3
39	5/5	5.3	−34.4	2.9	694.6	−84.5	240.4	3.3
40	6/6	355	−38.4	5.9	131.5	−83.0	351.3	6.6
42	5/5	6.5	−38.5	10	63.5	−82.3	258.1	10.2
43	6/6	336.2	−37.6	15.5	23.9	−67.3	21.0	9.9
44	5/5	347.5	−42.4	3.8	400	−75.9	1.4	4.6
46	6/6	338.9	−45.6	6.2	122.7	−68.0	5.4	8.3
48	5/5	357.7	−37	5.7	179.1	−85.5	334.8	6.1
61	6/6	346.2	−44.3	6.1	126.6	−74.2	358.8	7.8
62	6/6	352.1	−39	4.4	238.2	−80.9	0.2	5.4
67	7/7	359.2	−30.7	14.8	19	−88.5	35.0	15.0
68	4/6	353.1	−29.2	6.7	189.3	−83.3	46.1	6.3
71	4/6	359.1	−35.5	9.9	161.3	−87.0	325.3	5.0
**72	6/7	158.4	−40.3	15.1	23.5	43.0	302.9	63.8
73	5/5	353.7	−37.2	2.1	1293.5	−82.8	2.5	2.0
*78	6/6	8.3	−42.6	5	198	−78.9	264.1	6.1
*79	3/4	350.8	−40.8	10	153.4	−79.1	358.7	7.8
G-1	7/7	7.1	−40.6	8.8	47.7	−80.3	262.1	12.7
G-2	5/5	18.4	−43	13.5	33.3	−70.3	246.0	15.5
G-3	3/3	8.9	−40	7.4	277.8	−79.7	254.3	4.1
G-4	4/4	3.8	−37.5	12.7	53.6	−84.6	270.4	12.0
G-5	6/6	9.3	−39.7	8.8	59.1	−79.3	252.4	13.4
G-6	3/3	323.1	−29.1	7.8	249.8	−65.0	29.7	22.4

Notes: n N^{-1}, number of analysed specimens/number of specimens used in mean directions; α_{95}, Fisher's cone of confidence; *k*, Fisher's precision parameter; VGP, virtual geomagnetic pole; Plong, pole longitude; Plat, pole latitude; Swi, intrasite dispersion; Sites G-1 to G-6 correspond to groups of sites of Yokoyama *et al.* (2012); Site G-1 corresponds to sites AE1, AE3, AE5 and AE7; G-2 to sites AE10, AE 11 and AE 12; G-3 to sites AE 15 and AE 29; G-4 to sites AE 16, AE 20 and AE 25; G-5 to sites AE 23, AE 24 and AE 26; G-6 to sites AE 27 and AE 28.
*Sites of melt veins.
**Rejected site.

(IRM) acquisition curves and thermomagnetic curves. Hysteresis measurements were made using a MicroMag VSM (Princeton Instruments Corp.). IRM acquisitions were performed up to 2800 mT in more than 40 steps using a pulse magnetizer MMPM10 (Magnetic Measurements Ltd.) and a spinner magnetometer (Molspin Ltd.). Thermomagnetic curves were obtained through heating and cooling cycles from room temperature up to 700 °C in a CS3 furnace coupled with a KLY4S Kappabridge susceptometer (Agico). Experiments were done in Argon atmosphere to inhibit alteration during heating. All magnetic mineralogy studies were performed at the Paleomagnetic Laboratory of the University of São Paulo (USP).

RESULTS

For the impact melts, two magnetic carriers are indicated by thermal demagnetization patterns, thermomagnetic curves, hysteresis and IRM acquisition curves (Fig. 3). In thermal demagnetization (Fig. 3) and susceptibility versus temperature curves (Fig. 3), magnetic susceptibility typically decays steeply at about 580 °C (pure magnetite) followed by a continuous decrease until 700 °C (haematite). Accordingly, hysteresis loops are wasp-waisted (Fig. 3), typical of a mixture of two ferromagnetic phases with distinct coercivities (Tauxe *et al.* 1996). These two phases are also observed in IRM acquisition curves that show two saturation steps (Fig. 3), the first at fields below 1000 mT and the second at fields as high as 2500 mT. The AF demagnetization up to 140 mT effectively removed up to 80 per cent of the remanence for most samples. The remaining remanence was fully demagnetized after 680 °C (Fig. 4). Magnetic carriers of Araguainha granite rocks are magnetite and haematite (Yokoyama *et al.* 2012). According to Machado *et al.* (2009) and Yokoyama *et al.* (2012), these magnetic carriers were formed instantly at the beginning of the cratering process at the expense of biotite during incongruent melting induced by the impact. Thickness of melt bodies in Araguainha vary from 1 mm (thin melt veins in granite) to 400 m (thickness of the melt sheet), providing in consequence a wide range of cooling times thus reinforcing the idea that SV may have been averaged out during remanence acquisition by the Araguainha melts and granitic basement.

From the 169 analysed specimens, 140 presented a single stable magnetic direction, carried by both magnetite and haematite, with a negligible viscous overprint (Fig. 4). Some secondary directions correspond to coercivities until 20 mT and maximum unblocking temperatures of 350 °C. For 20 specimens (nine sites), this secondary direction was not completely isolated from the ChRM by

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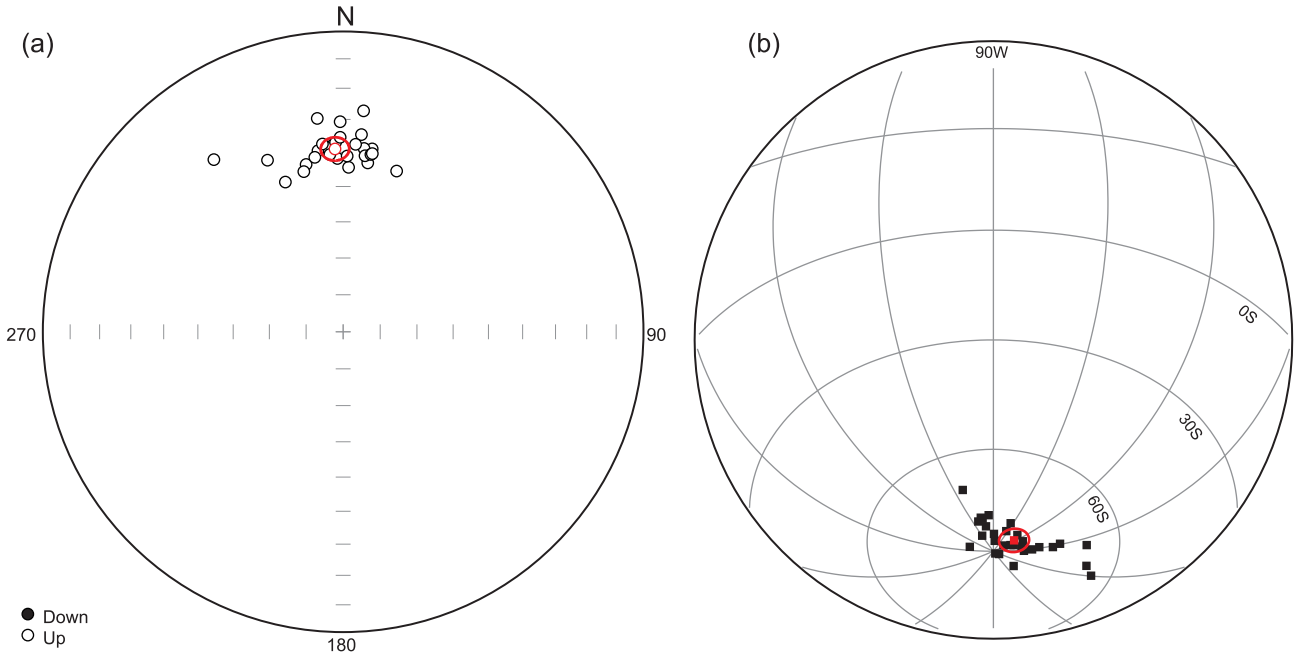


Figure 5. Mean directions and for the Araguainha collection (in red): (a) site mean characteristic directions and (b) site mean virtual geomagnetic poles.

vectorial subtraction. In this case, they were analysed by great circles (Halls 1976). PCA for vectors and great circles presented maximum angular deviation (MAD) below 10°.

Mean directions for each site are presented in Table 1 and Fig. 5. Directions are always well-grouped as attested by k values higher than 100 for most sites (Table 1). The Araguainha ChRMs are all of normal polarity, compatible with the magnetization being acquired after the Permo-Carboniferous Reversed Superchron (PCRS), which ended ca. 269 Ma (Lanci *et al.* 2013), and the secondary direction found is random. Before defining the palaeomagnetic pole, the cut-off method of Vandamme (1994) was applied for data selection. The final cut-off angle was 20.6° after rejection of only one data (site 72), which corresponds to the highest intrasite dispersion (Swi = 63.8°). The dispersion parameter of the mean palaeomagnetic pole was $S_B = 9.6^{\circ}{}^{+11.8}_{-8.1}$ (Fig. 6). The final palaeomagnetic pole of AIS falls at Lat = −84.2°; Lon = 326.6° ($K = 60$; $A_{95} = 3.6^{\circ}$, $N = 27$).

The dispersion of site-based palaeomagnetic directions was evaluated to test whether sampling had averaged out the SV, which generally increases with latitude. We use the dispersion pole parameter (S_B) of Biggin *et al.* (2008), which takes into account only the SV and eliminates variations associated with experimental errors or intrasite dispersion. Unfortunately, there are no models of SV dispersion for Permo-Triassic, so we compared our results to Biggin *et al.* (2008) models for the Jurassic and Cretaceous Normal Superchron (CNS), which represent extremes of reversal rate (Fig. 6). Moreover, we compared our data with the SV data of equatorial red bed sections from Dôme de Barrot and Lodeve (Haldan *et al.* 2009). Our data plot just below model estimates of PSV for Jurassic rapidly reversing geomagnetic field and within the lower range of S_B values for the CNS (Fig. 6). These results are similar to the sedimentary units studied by Haldan *et al.* (2009) which were deposited during the Permo-Carboniferous Reversal Superchron (PCRS). Since the dispersion parameter obtained for Araguainha is comparable to that obtained for units similar in age and the thermal model suggests a reasonable cooling time for impact melts, it is likely that the palaeo SV has been averaged out (almost completely) by the impact melt and the granitic basement of Araguainha.

DISCUSSION

Age of the Araguainha impact

Geochronological data obtained from multiminerals using different dating techniques have constrained the age of the Araguainha impact melt to 254.7 ± 2.5 Ma, close to the Permo-Triassic boundary (Tohver *et al.* 2012). In order to refine the age interval in which the Araguainha impact event occurred, we compared our palaeomagnetic data with the global polarity timescale (GPTS). All magnetically stable samples of the impact melt and granite record a unique normal geomagnetic polarity. Though there is still considerable discussion as to the precise age of individual chrons from this time period (*cf.* review by Steiner 2006), the coarse pattern of reversals is fairly robust. Examining the latest GPTS of Gradstein *et al.* (2012) reveals a predominance of normal polarity chrons towards the younger portion of this interval (ca. 252–256 Ma), whereas reversed polarity seems more prevalent for the >256 Ma period (Fig. 7). Although a robust use of the Araguainha magnetization polarity awaits a more finely tuned GPTS for the latest Permian, we suggest that the age of the impact event is <256 Ma, consistent with the younger portion of the 254.7 ±2.5 Ma range provided by geochronological study of the impact melt and granite.

South American APW paths

The relative positions of Laurussia and Gondwana within the Pangaean supercontinent are reconstructed from the APW paths (APWPs) of these two continents. There are two contrasting proposals for the APWP for South America for the Carboniferous to Triassic interval that makes exclusive use of data from South America (Fig. 8). The first path (purple in Fig. 8) mostly comprises data

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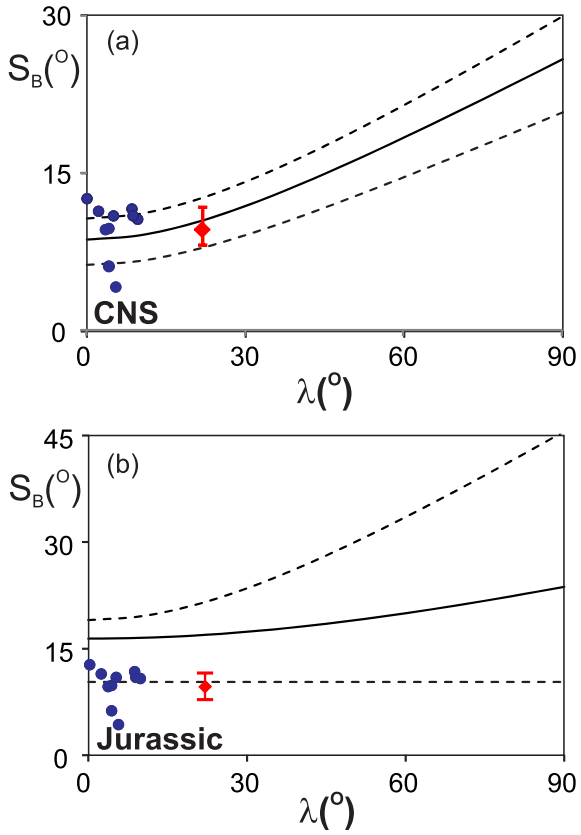


Figure 6. Comparison of S-parameter with (a) Cretaceous Normal Superchron (CNS) and (b) Jurassic models (from Biggin *et al.* 2008), solid blue circles represent the *S*-parameter for Permo-Carboniferous Reversed Superchron (PCRS) from Haldan *et al.* (2009).

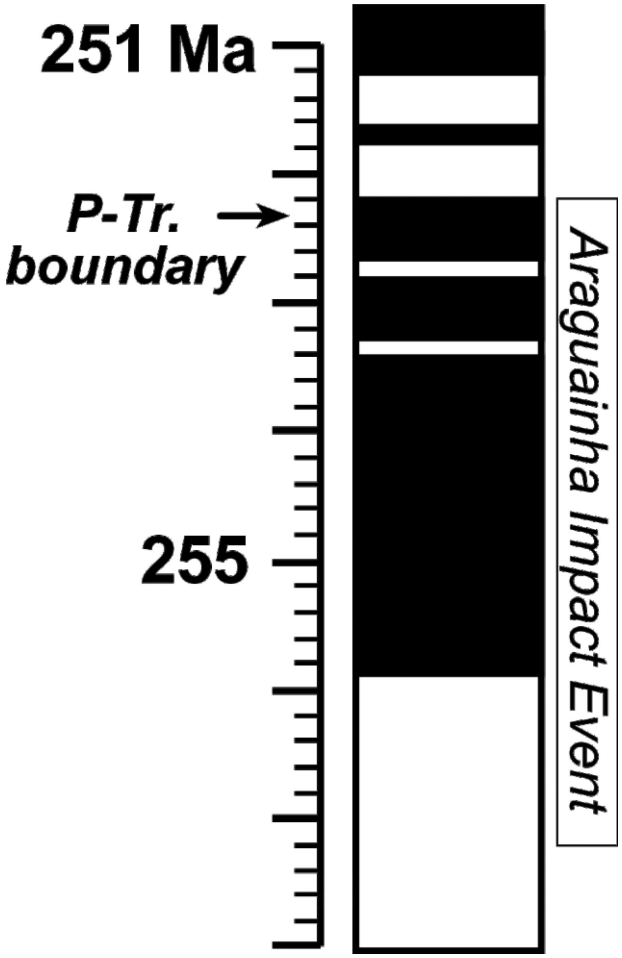


Figure 7. Synthetic reversal column from Gradstein *et al.* (2012) with the Araguainha melt sheet age interval superimposed.

from sedimentary rocks (e.g. Rapalini *et al.* 2006; Tomezzoli 2009; Tomezzoli *et al.* 2013). This path is marked by a cusp in late Permian times, and implies a more northerly palaeolatitude for Gondwana in late Permian times, effectively worsening the continental overlap that underpins the Pangea-B hypothesis. The second path (grey in Fig. 8) is constructed from a data set that excludes poles on the basis of suspected remagnetization or demonstrably weak, antiquated data sets (e.g. Brandt *et al.* 2009; Domeier *et al.* 2012). This path is shorter than the first one between the Early-Permian (eP) and the Early-Triassic (eTr) and excludes the late Permian cusp. It is generally more compatible with the Pangea-A hypothesis.

Much of the difference in the length and shape of the two paths can be explained by systematic biases in the record of the geomagnetic field from study of sedimentary rocks. Palaeomagnetic remanence in sedimentary rocks is vulnerable to biases caused by inclination shallowing and remagnetization events (e.g. Tauxe & Kent 1984; Bilardello & Kodama 2010; Font *et al.* 2012). In addition, few of the palaeomagnetically studied sedimentary sequences from South America have geochronological age constraints, or are adequately sampled to account for SV. In the first path, the age of some poles was not directly determined but inferred from their location on the APW path (Fig. 8). A clear example is provided by the Copacabana Group pole reported by Rakotosolofo *et al.* (2006). These rocks are Early Permian (Asselian-Sakmarian) according to palaeontological and palynological studies (De la Cruz *et al.* 1998), they are reversely magnetized consistent with deposition during the Permo-Carboniferous Reverse Superchron, and the characteristic remanence is primary, based on a positive fold test at the 99 per cent confidence level. However, Tomezzoli (2009) has assigned a Triassic age for the pole based only on its position on the first path.

Implications for Pangea configurations

The most important effect contributing to the elongated first path seems to be the inclination shallowing of magnetic directions due to vertical compaction since most reference poles for the Carboniferous to Triassic were obtained from sedimentary units (e.g. Bilardello & Kodama 2010; Domeier *et al.* 2012). Some palaeomagnetic poles on both sides of Pangea were corrected for their potential shallowing in inclination using different methods, with flattening factors varying from 1.0 to 0.5 (e.g. Rapalini *et al.* 2006; Brandt *et al.* 2009; Bilardello & Kodama 2010). In addressing this problem, Domeier *et al.* (2012) has applied a single flattening factor of 0.6 to all Pangean sedimentary units that were not corrected in the original study (Figs 9a and b). Applying this correction brings sedimentary poles close to the coeval igneous poles (Figs 9a and b), suggestive of the systematic effect of inclination shallowing in the sedimentary units.

Fig. 9 shows the Araguainha pole (AIS) together with other palaeomagnetic poles of West Gondwana from the compilation of Domeier *et al.* (2012). This figure shows palaeomagnetic poles from 270 to 240 Ma, which are derived from the study of both sedimentary (white and grey) and volcanic rocks (red). Igneous-based poles are

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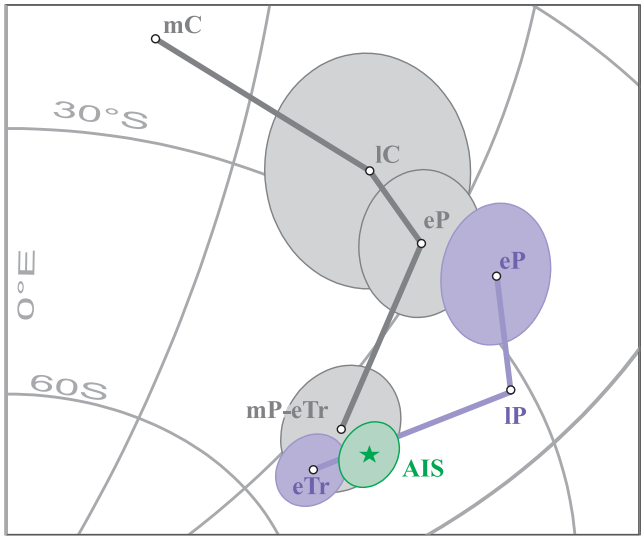


Figure 8. Different APW paths proposed for South America (first path in purple by Tomezzoli 2009; second path in grey by Brandt *et al.* 2009). The paths span the middle Carboniferous (mC), late Carboniferous (lC), early Permian (eP), middle Permian (mP), late Permian (lP) and early Triassic (eTr). The Araguainha pole (AIS) with its confidence error is indicated in green.

very scarce after 270 Ma, but the available poles are more clustered than those obtained for sedimentary rocks. The ca. 254 Ma AIS pole falls within the igneous group, close to the new 263 Ma Sierra Chica pole from the Colorado volcanic province of Argentina (Domeier *et al.* 2011), suggesting that the latter area is structurally coherent with Gondwana. Furthermore, these two igneous poles provide clear well-dated reference poles that are less likely to suffer from the systematic biases that afflict sedimentary rocks. Recalculated mean poles for West Gondwana for 260 and 250 Ma incorporating the new AIS result to the data set of Domeier *et al.* (2012) demonstrate some slight improvement in A_{95} confidence intervals (Fig. 10).

CONCLUSIONS

A systematic palaeomagnetic study was carried out on the 254.7 ± 2.5 Ma AIS (Central Brazil), which affected the basement and the sedimentary pile of the Paraná basin at the stable interior of the Gondwanan supercontinent. The normal magnetization disclosed for the impact melts and the granitic host is consistent with a late Permian to early Triassic age for the impact event, given the preponderance of normal polarity magnetozones from this time period. The analysis of palaeomagnetic dispersion suggests that most if not all SV was averaged out. Comparison of the Araguainha palaeomagnetic pole (AIS) with igneous and sedimentary palaeomagnetic poles from middle Permian to early Triassic South America reinforces the observation that sedimentary poles are biased to shallow inclinations. Restricting palaeogeographic interpretations to igneous poles such as that obtained from the Araguainha melts reduces the overlap of Gondwana with Laurussia in a Pangea-A configuration.

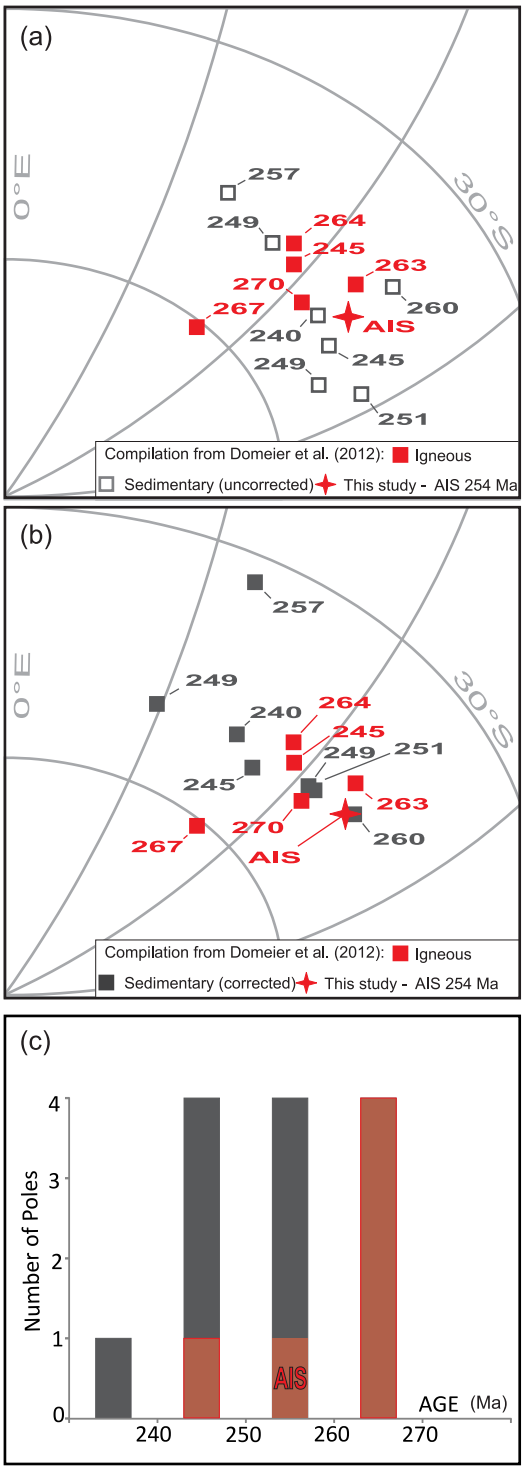


Figure 9. Comparison between igneous (red) and sedimentary (white and grey) 270–240 Ma poles from West Gondwana: (a) before shallowing correction; (b) after a uniform correction of $f = 0.6$ and (c) the histogram shows a lack of igneous poles for this period of time.

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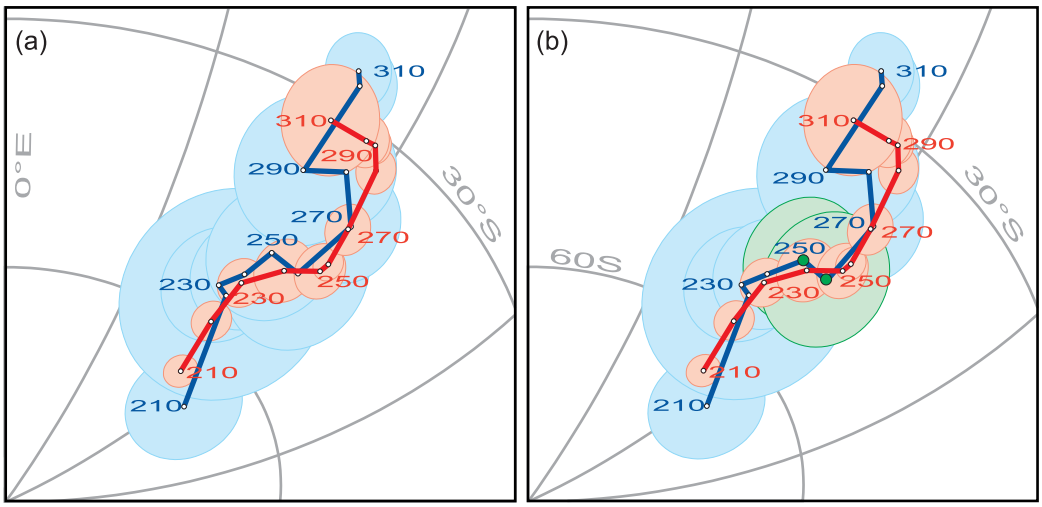


Figure 10. Modified 310–200 Ma APWP segment of West Gondwana: (a) original APWP as defined by Domeier *et al.* (2012); (b) modified APWP including the AIS pole. Blue curves and blue ellipses represent the South America APWP and the poles confidence error ellipses, and red curves and red ellipses represent the Laurussia APWP and the poles confidence error ellipses, respectively. The green circles and their respective confidence error ellipses represent the mean poles recalculated (this study).

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